



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
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(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of CSE

"A place to learn; A chance to grow"

2025-2026(Odd Semester)

MSE-1 EXAMINATION

Course: B.Tech. In Computer Science and Engineering

Subject: LLM&PE

Max Marks: 20

Date: 12/08/2025

Year/Semester: 5th Semester (3rd Year)

Subject Code: CS5M003

Duration:- 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

- CO1: Students will be able to Understand and apply basic and advanced prompt engineering techniques.
CO2: Students will be able to Integrate and utilize APIs to build AI applications.
CO3: Students will be able to Design and implement AI-powered chatbots and utilities.
CO4: Students will be able to Optimize generative AI applications for latency and cost.
CO5: Students will be able to Evaluate prompt performance and select appropriate fallback strategies.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	In what way does a system message help guide the output of an AI model?	L2/CO1	1
Q.2	Identify the type of prompting used in the following instruction: "Translate the sentence into Hindi: 'How are you?'"	L1/CO1	1
Q.3	Give the example to show how temperature parameter control in a language model?	L3/CO2	1
Q.4	Identify the prompting pattern used in the following instruction: "Let's solve this step-by-step."	L1/CO2	1

SECTION B

1. All Questions are compulsory.
2. Answer in one two three sentence.

Q.5	Explain the role of context in a prompt with a simple example.	L2/CO1	2
Q.6	Apply your knowledge to construct a one-shot prompt to convert temperatures from Celsius to Fahrenheit.	L3/CO1	2
Q.7	Explain the role of the top-p parameter in prompt engineering.	L2/CO2	2
Q.8	Apply your knowledge to construct a prompt to generate a short story using a specific persona of a humorous children's book author.	L3/CO2	2

SECTION C

1. Solve any two questions.

Q.9	Analyze the following prompt and identify the missing components. Then, rewrite it as a complete and effective prompt using proper structure.	L4/CO1	4
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Translate this into spanish - "Good morning"

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| Q.10 | You are developing a customer support chatbot using an LLM. Apply appropriate advanced prompt engineering techniques (such as persona creation, temperature tuning, and prompt design patterns) to ensure that the chatbot provides helpful, friendly, and consistent responses. Illustrate with an example prompt. | L3/CO2 | 4 |
| Q.11 | Analyze the concept of prompt design patterns used in advanced prompt engineering. Choose any two patterns and explain, with examples, how each pattern improves the reasoning or response quality of a language model. | L4/CO2 | 4 |